

Defining a Function

Here's a simple function named `greet_user()` that prints a greeting:

```
greeter.py def greet_user():
            """Display a simple greeting."""
            print("Hello!")

greet_user()
```

This example shows the simplest structure of a function. The first line uses the keyword `def` to inform Python that you're defining a function. This is the *function definition*, which tells Python the name of the function and, if applicable, what kind of information the function needs to do its job. The parentheses hold that information. In this case, the name of the function is `greet_user()`, and it needs no information to do its job, so its parentheses are empty. (Even so, the parentheses are required.) Finally, the definition ends in a colon.

Any indented lines that follow `def greet_user()`: make up the *body* of the function. The text on the second line is a comment called a *docstring*, which describes what the function does. When Python generates documentation for the functions in your programs, it looks for a string immediately after the function's definition. These strings are usually enclosed in triple quotes, which lets you write multiple lines.

The line `print("Hello!")` is the only line of actual code in the body of this function, so `greet_user()` has just one job: `print("Hello!")`.

When you want to use this function, you have to call it. A *function call* tells Python to execute the code in the function. To *call* a function, you write the name of the function, followed by any necessary information in parentheses. Because no information is needed here, calling our function is as simple as entering `greet_user()`. As expected, it prints `Hello!`:

```
Hello!
```

Passing Information to a Function

If you modify the function `greet_user()` slightly, it can greet the user by name. For the function to do this, you enter `username` in the parentheses of the function's definition at `def greet_user()`. By adding `username` here, you allow the function to accept any value of `username` you specify. The function now expects you to provide a value for `username` each time you call it. When you call `greet_user()`, you can pass it a name, such as `'jesse'`, inside the parentheses:

```
def greet_user(username):
    """Display a simple greeting."""
    print(f"Hello, {username.title()}!")

greet_user('jesse')
```
